ID: 87023f34

"Ghosts of the Old Year" is an early 1900s poem by James Weldon Johnson. In the poem, the speaker describes experiencing an ongoing cycle of anticipation followed by regretful reflection: _____

Which quotation from "Ghosts of the Old Year" most effectively illustrates the claim?

A. "What does this brazen tongue declare, / That falling on the midnight air / Brings to my heart a sense of care / Akin to fright?"

B. "The snow has ceased its fluttering flight, / The wind sunk to a whisper light, / An ominous stillness fills the night, / A pause—a hush."

C. "It tells of many a squandered day, / Of slighted gems and treasured clay, / Of precious stores not laid away, / Of fields unreaped."

D. "And so the years go swiftly by, / Each, coming, brings ambitions high, / And each, departing, leaves a sigh / Linked to the past."

ID: 73d457b6

In the 1970s, a roughly 60,000-year-old piece of hyena bone marked with nine notches was discovered at a site in western France once inhabited by Neanderthals. Although many believe that only modern humans developed systems for notating numbers, one archaeologist asserts that this artifact may be a sign that Neanderthals also recorded numerical information. The notches on the bone are unevenly spaced but approximately parallel, and microscopic analysis reveals that they were made with a single stone tool; according to the archaeologist, this suggests that the notches were all made at one time by one individual as a means of counting something.

Which finding, if true, would most directly weaken the underlined claim?

A. Parallel lines are a common feature in modern humans' early systems for recording numerical information.

B. More than nine approximately parallel notches made with a different stone tool are present on another artifact found at a site in western France.

C. It would have taken careful effort to make evenly spaced lines on bone with the stone tools typically used by Neanderthals.

D. Decorative art discovered at another Neanderthal site in western France primarily features patterns of unevenly spaced parallel lines.

ID: b4cda84d

In 1967 the US Congress created the Corporation for Public Broadcasting, which in turn created National Public Radio (NPR). NPR began producing and distributing high-quality news and cultural programming to affiliate stations across the United States in 1971. In a research paper, a student claims that the Corporation for Public Broadcasting and NPR were inspired by the British Broadcasting System (BBC), which had been established in the 1920s.

Which quotation from a work by a historian would be the most effective evidence for the student to include in support of this claim?

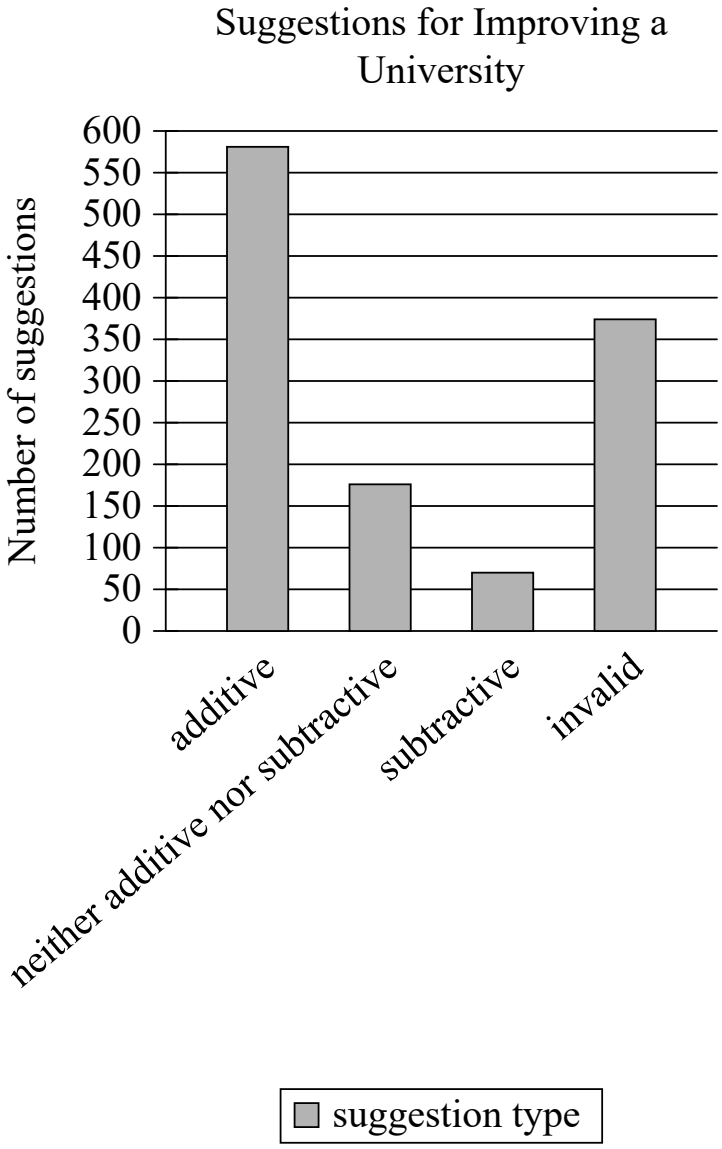
A. "Although the BBC had begun as a private corporation, politicians successfully argued to make it a public company because they believed a public broadcaster could help build national unity in the aftermath of World War I."

B. "For many decades, the BBC had no competition since it held Britain's only broadcasting license, whereas in the United States, the Corporation for Public Broadcasting launched NPR in a broadcasting market already filled with competitors."

C. "Congress's embrace of publicly funded broadcasting reflected a common belief among US politicians that the role of government was not only to ensure people's safety and liberty but also to enrich people's lives in other ways."

D. "The goal of the BBC was to support British democracy by promoting an informed citizenry, and US legislators believed that ensuring access to high-quality programming could do the same for democracy in the United States."

ID: 23e2421a



Gabrielle Adams and colleagues reviewed suggestions for improving a university that had been submitted to the university's president. They coded each suggestion as additive (the idea suggested adding something new to the university), subtractive (the idea suggested removing something from the university), neither additive nor subtractive, or invalid (the idea was not comprehensible). The data illustrated people's tendency to overlook the possibility of removing things to achieve improvements: _____

Which choice most effectively uses data in the graph to complete the statement?

A. around 175 suggestions were coded as neither additive nor subtractive, whereas around 575 suggestions were coded as additive.

B. more than 350 suggestions were coded as invalid, whereas fewer than 100 suggestions were coded as subtractive.

C. fewer than 100 suggestions were coded as subtractive, whereas more than 550 suggestions were coded as additive.

D. around 575 suggestions were coded as additive, whereas around 175 suggestions were coded as subtractive.

ID: dc47c2ac

Although most songbirds build open, cupped nests, some species build domed nests with roofs that provide much more protection. Many ecologists have assumed that domed nests would provide protection from weather conditions and thus would allow species that build them to have larger geographic ranges than species that build open nests do. To evaluate this assumption, a research team led by evolutionary biologist Iliana Medina analyzed data for over 3,000 species of songbirds.

Which finding from Medina and her colleagues' study, if true, would most directly challenge the assumption in the underlined sentence?

A. Species that build open nests tend to have higher extinction rates than species that build domed nests.

B. Species that build open nests tend to be smaller in size than species that build domed nests.

C. Species that build open nests tend to use fewer materials to build their nests than species that build domed nests do.

D. Species that build open nests tend to have larger ranges than species that build domed nests.

ID: 84136d69

Five of the Responses to Survey about Actions to Conserve Energy

Action	Action category	Percentage of respondents selecting action (%)
Use efficient cars/hybrids	efficiency	2.8
Change thermostat setting	curtailment	6.3
Use bike or public transportation instead of car	curtailment	12.9
Use efficient light bulbs	efficiency	3.6
Turn off lights	curtailment	19.6

In a survey of public perceptions of energy use, researcher Shahzeen Attari and her team asked respondents to name the most effective action ordinary people can take to conserve energy. The team categorized each action as either an efficiency or a curtailment and found that respondents tended to name curtailments more often than they did efficiencies. For example, 19.6% of respondents stated that the most effective way to conserve energy is to turn off the lights, while only _____

Which choice most effectively uses data from the table to complete the text?

A. 6.3% of respondents said it was most effective to use efficient cars or hybrids.

B. 2.8% of respondents said it was most effective to change the thermostat setting.

C. 12.9% of respondents said it was most effective to use a bike or public transportation.

D. 3.6% of respondents said it was most effective to use efficient light bulbs.

ID: 9aa5efc4

Born in 1891 to a Quechua-speaking family in the Andes Mountains of Peru, Martín Chambi is today considered to be one of the most renowned figures of Latin American photography. In a paper for an art history class, a student claims that Chambi's photographs have considerable ethnographic value—in his work, Chambi was able to capture diverse elements of Peruvian society, representing his subjects with both dignity and authenticity.

Which finding, if true, would most directly support the student's claim?

A. Chambi took many commissioned portraits of wealthy Peruvians, but he also produced hundreds of images carefully documenting the peoples, sites, and customs of Indigenous communities of the Andes.

B. Chambi's photographs demonstrate a high level of technical skill, as seen in his strategic use of illumination to create dramatic light and shadow contrasts.

C. During his lifetime, Chambi was known and celebrated both within and outside his native Peru, as his work was published in places like Argentina, Spain, and Mexico.

D. Some of the peoples and places Chambi photographed had long been popular subjects for Peruvian photographers.

ID: 16a4a83b

An Ideal Husband is an 1895 play by Oscar Wilde. In the play, which is a satire, Wilde suggests that a character named Lady Gertrude Chiltern is perceived as both extremely virtuous and unforgiving, as is evident when another character says _____

Which quotation from *An Ideal Husband* most effectively illustrates the claim?

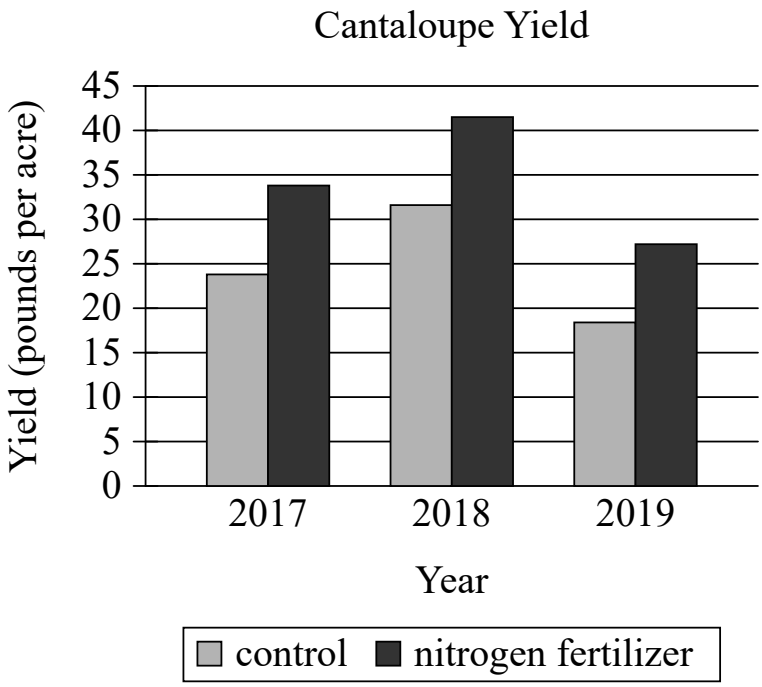
A. "Lady Chiltern is a woman of the very highest principles, I am glad to say. I am a little too old now, myself, to trouble about setting a good example, but I always admire people who do."

B. "Do you know, [Lady Chiltern], I don't mind your talking morality a bit. Morality is simply the attitude we adopt towards people whom we personally dislike."

C. "[Lady Chiltern] does not know what weakness or temptation is. I am of clay like other men. She stands apart as good women do—pitiless in her perfection—cold and stern and without mercy."

D. "Lady Chiltern, you are a sensible woman, the most sensible woman in London, the most sensible woman I know."

ID: 1db1a9a6



To test the effects of a nitrogen fertilizer on cantaloupe production, researchers grew cantaloupe plants and harvested their fruit over three years. In each year, half the plants were grown using a nitrogen fertilizer, and the other half were grown using a control fertilizer that contained no nitrogen. The researchers concluded that the nitrogen fertilizer increases cantaloupe yield.

Which choice best describes data in the graph that support the researchers' conclusion?

A. In every year of the experiment, plants treated with the nitrogen fertilizer had a yield of at least 30 pounds per acre.

B. In every year of the experiment, plants treated with the nitrogen fertilizer had a greater yield than did plants treated with the control fertilizer.

C. The 2018 yield for plants treated with the control fertilizer was greater than was the 2019 yield for plants treated with the nitrogen fertilizer.

D. The yield for plants treated with the nitrogen fertilizer increased from 2017 to 2018.

ID: dd349efc

Participants' Evaluation of the Likelihood That Robots Can Work Effectively in Different Occupations

Occupation	Somewhat or very unlikely (%)	Neutral (%)	Somewhat or very likely (%)
television news anchor	24	9	67
teacher	37	16	47
firefighter	62	9	30
surgeon	74	9	16
tour guide	10	8	82

Rows in table may not add up to 100 due to rounding.

Georgia Tech roboticists De'Aira Bryant and Ayanna Howard, along with ethicist Jason Borenstein, were interested in people's perceptions of robots' competence. They recruited participants and asked them how likely they think it is that a robot could do the work required in various occupations. Participants' evaluations varied widely depending on which occupation was being considered; for example, _____

Which choice most effectively uses data from the table to complete the example?

A. 47% of participants believe that it is somewhat or very likely that a robot could work effectively as a teacher, but 37% of respondents believe that it is somewhat or very unlikely that a robot could do so.

B. 9% of participants were neutral about whether a robot could work effectively as a television news anchor, which is the same percent of participants who were neutral when asked about a robot working as a surgeon.

C. 82% of participants believe that it is somewhat or very likely that a robot could work effectively as a tour guide, but only 16% believe that it is somewhat or very likely that a robot could work as a surgeon.

D. 62% of participants believe that it is somewhat or very unlikely that a robot could work effectively as a firefighter.

ID: 1d08c7ee

Pulitzer Prize–winning writer Héctor Tobar has built a multifaceted career as both a journalist and an author of short stories and novels. In an essay about Tobar's work, a student claims that Tobar blends his areas of expertise by applying journalism techniques to his creation of works of fiction.

Which quotation from a literary critic best supports the student's claim?

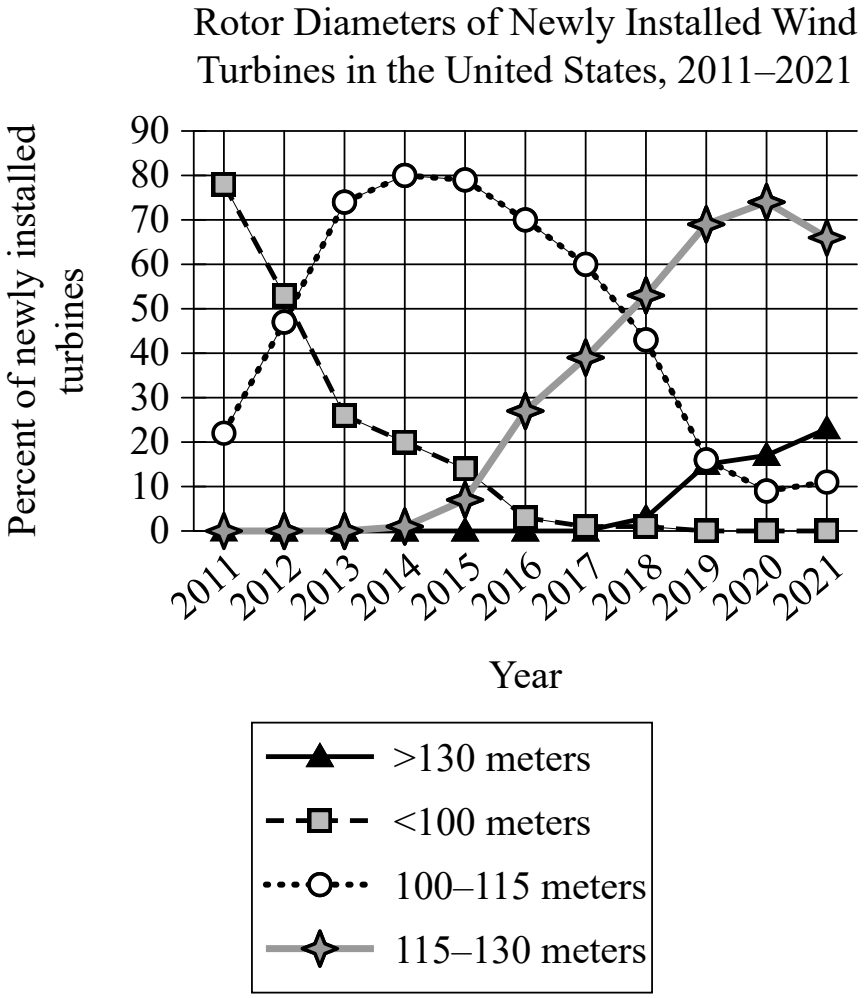
A. "For one novel, an imagined account of a real person's global travels, Tobar approached his subject like a reporter, interviewing people the man had met along the way and researching the man's own writings."

B. "Tobar got his start as a volunteer for *El Tecolote*, a community newspaper in San Francisco, and wrote for newspapers for years before earning a degree in creative writing and starting to publish works of fiction."

C. "Many of Tobar's notable nonfiction articles are marked by the writer's use of techniques usually associated with fiction, such as complex narrative structures and the incorporation of symbolism."

D. "The protagonist of Tobar's third novel is a man who wants to be a novelist and keeps notes about interesting people he encounters so he can use them when developing characters for his stories."

ID: 2df730d0



All other things being equal, the larger a wind turbine's rotor diameter (the diameter of the imaginary circle swept by the turbine's rotating blades), the greater amount of energy the turbine can generate. In a research paper on wind power, a student claims that in the United States, the amount of energy generated per newly installed turbine increased substantially between 2011 and 2021.

Which choice best describes data in the graph that support the student's claim?

A. The percentage of newly installed turbines with rotor diameters greater than 130 meters increased every year between 2011 and 2021.

B. In 2011, nearly 80% of turbines installed had rotor diameters of less than 100 meters, whereas only a little more than 20% of turbines installed that year had rotor diameters of 100–115 meters.

C. No turbines installed in 2011 had rotor diameters greater than 115 meters, whereas the majority of turbines installed in 2021 had rotor diameters greater than 130 meters.

D. Most turbines installed in 2011 had rotor diameters of less than 100 meters, whereas most turbines installed in 2021 had rotor diameters of at least 115 meters.

ID: 5d6ab069

Jan Gimsa, Robert Sleigh, and Ulrike Gimsa have hypothesized that the sail-like structure running down the back of the dinosaur *Spinosaurus aegyptiacus* improved the animal's success in underwater pursuits of prey species capable of making quick, evasive movements. To evaluate their hypothesis, a second team of researchers constructed two battery-powered mechanical models of *S. aegyptiacus*, one with a sail and one without, and subjected the models to a series of identical tests in a water-filled tank.

Which finding from the model tests, if true, would most strongly support Gimsa and colleagues' hypothesis?

A. The model with a sail took significantly less time to complete a sharp turn while submerged than the model without a sail did.

B. The model with a sail displaced significantly more water while submerged than the model without a sail did.

C. The model with a sail had significantly less battery power remaining after completing the tests than the model without a sail did.

D. The model with a sail took significantly longer to travel a specified distance while submerged than the model without a sail did.

ID: ccf414c9

E-book Sales as a Percentage of Total Unit Sales in All Book Formats for a Large US Trade Publisher, by Genre, 2006, 2011, 2016

Genre	2006	2011	2016
science fiction and fantasy	0.6	27.7	36.7
cookbooks	0	2.9	10.5
travel guides	0	5.5	24.6
romance	0.3	40.6	56.2

E-books became an increasingly popular means of reading in the United States in the 2000s and 2010s, though that popularity was concentrated in titles that, like those in most fiction genres, are meant to be read straight through from beginning to end. For books in nonfiction genres that do not tell stories and require the reader to flip back and forth through a volume, e-books were significantly less commercially successful. This can be seen by comparing _____

Which choice most effectively uses data from the table to illustrate the claim?

A. the percentage of 2016 cookbook sales that were e-books with the percentage of 2016 science fiction and fantasy sales that were e-books.

B. the percentage of 2006 romance sales that were e-books with the percentage of 2016 romance sales that were e-books.

C. the percentage of 2006 romance sales that were e-books with the 2006 science fiction and fantasy sales that were e-books.

D. the percentage of 2011 travel guide sales that were e-books with the percentage of 2016 travel guide sales that were e-books.

ID: c6b470bb

"Odalie" is an 1899 short story by Alice Dunbar-Nelson. In the story, a young woman named Odalie attends the annual Mardi Gras carnival in New Orleans, where she lives with her guardian Tante Louise. Dunbar-Nelson portrays Odalie as eager to escape the monotony of her everyday life: _____

Which quotation from "Odalie" most effectively illustrates the claim?

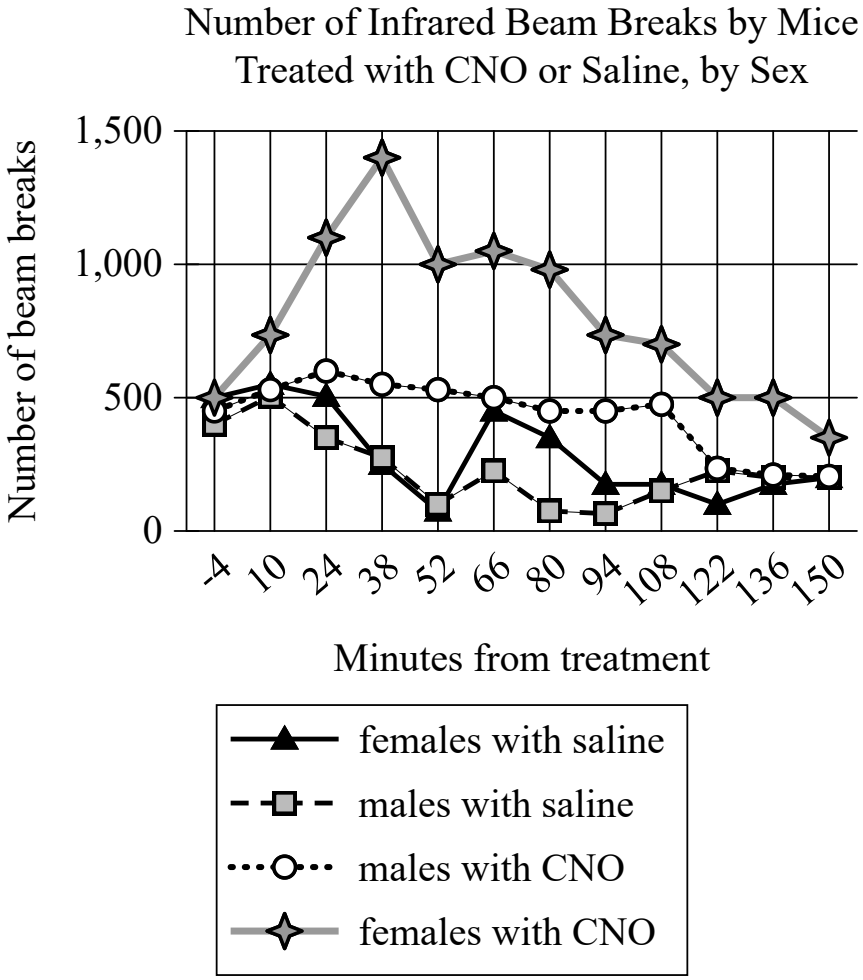
A. "Mardi Gras was a tiresome day, after all, she sighed, and Tante Louise agreed with her for once."

B. "In the old French house on Royal Street, with its quaint windows and Spanish courtyard green and cool, and made musical by the plashing of the fountain and the trill of caged birds, lived Odalie in convent-like seclusion."

C. "When one is shut up in a great French house with a grim sleepy tante and no companions of one's own age, life becomes a dull thing, and one is ready for any new sensation."

D. "It was Mardi Gras day at last, and early through her window Odalie could hear the jingle of folly bells on the [participants'] costumes, the tinkle of music, and the echoing strains of songs."

ID: e2829dd7



To investigate the influence of certain estrogen-responsive neurons on energy expenditure, biologist Stephanie Correa et al. treated female and male mice with either saline solution or clozapine-N4-oxide (CNO), which activates the neurons. Monitoring the activity levels of the mice by measuring how frequently the animals broke infrared beams crossing their enclosures, Correa et al. found that the mice in their study showed sex-specific differences in response to neuron activation: _____

Which choice most effectively uses data from the graph to complete the assertion?

A. the four groups of mice differed greatly in their activity levels before treatment but showed identical activity levels at the end of the monitoring period.

B. saline-treated females showed substantially more activity at certain points in the monitoring period than saline-treated males did.

C. CNO-treated females showed more activity relative to saline-treated females than CNO-treated males showed relative to saline-treated males.

D. CNO-treated females showed a substantial increase and then decline in activity over the monitoring period, whereas CNO-treated males showed a substantial decline in activity followed by a steep increase.

ID: cbecb873

Body Length, Filter Time, and Lunges per Dive for Four Whale Species

Whale species	Typical adult body length (meters)	Average time to filter all engulfed water (seconds)	Average number of lunges per dive deeper than 50 meters
fin	18–22	31.30	3.95
humpback	11–17	17.12	6.28
minke	7–10	8.88	7.48
blue	24–34	60.27	4.02

Some whale species practice lunge feeding, in which they lunge toward prey with their mouths open at wide angles, collect the prey and the surrounding water, and then filter out the water through baleen plates in their mouths. Although the volume of water engulfed increases with whales' body length, the surface area of whales' baleen plates, which influences the rate at which water can be filtered, does not increase with body length to the same degree, which helps explain why _____

Which choice most effectively uses data from the table to complete the statement?

A. minke whales and humpback whales show similar average filter times.

B. humpback whales show an average of 6.28 lunges per dive.

C. fin whales show a longer average filter time than minke whales do.

D. blue whales show the longest average filter time and the highest average number of lunges per dive.

ID: 25290c8d

"On Virtue" is a 1766 poem by Phillis Wheatley. Wheatley addresses the poem directly to the quality of virtue, imploring it to assist her in reaching a future goal: _____

Which quotation from "On Virtue" most effectively illustrates the claim?

A. "Attend me, *Virtue*, thro' my youthful years! / O leave me not to the false joys of time! / But guide my steps to endless life and bliss."

B. "I cease to wonder, and no more attempt / Thine height t'explore, or fathom thy profound."

C. "O thou bright jewel in my aim I strive / To comprehend thee. Thine own words declare / Wisdom is higher than a fool can reach."

D. "But, O my soul, sink not into despair, / *Virtue* is near thee, and with gentle hand / Would now embrace thee, hovers o'er thine head."

ID: 2584bcfb

Matthew D. Rocklage and team examined whether consumer ratings of movies can predict box office success. The team considered both numeric star ratings and written reviews in their research. To analyze the written reviews, the team measured the emotionality—the degree to which a written review expresses an emotional reaction—of user reviews on a movie rating website, assigning each review an emotionality score. After reviewing this research, a student argues that the emotionality of movie reviews is unrelated to a movie's success at the box office.

Which finding, if true, would most directly weaken the student's conclusion?

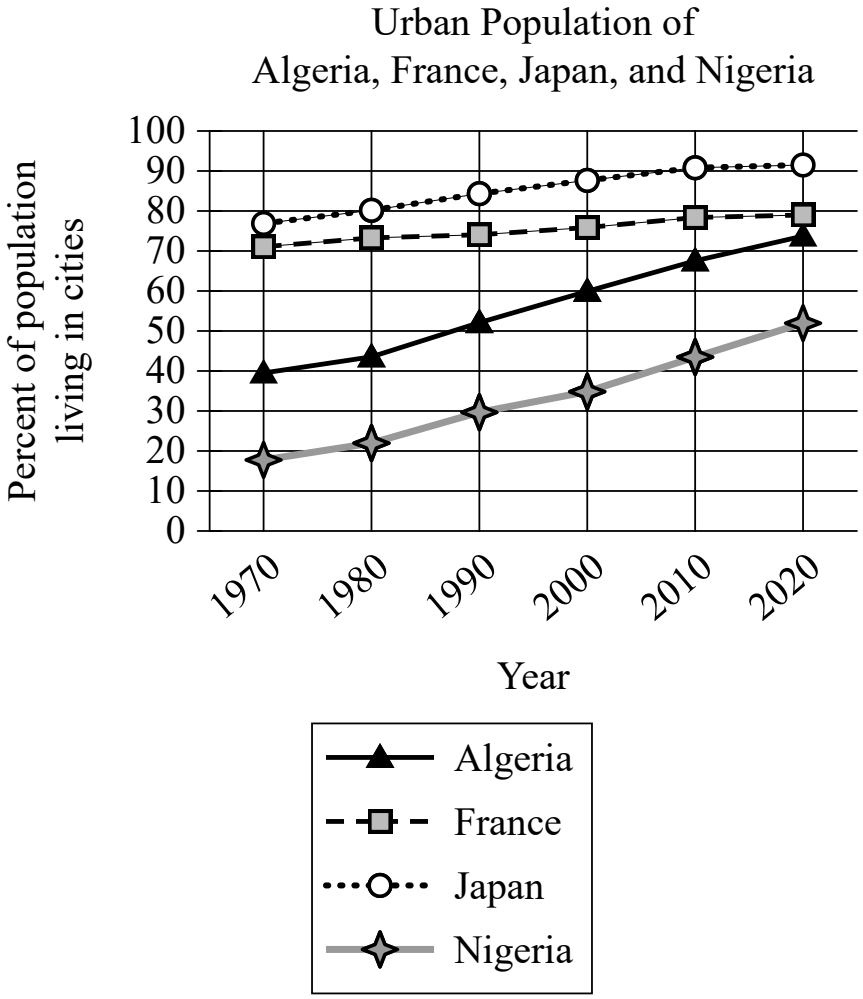
A. Movies that had the highest average emotionality scores received the lowest average star ratings on the movie rating website.

B. The average emotionality score of a movie's reviews was a positive predictor of that movie's box office earnings.

C. More than half of the movies that the team examined received an average star rating of 3 out of 5 stars.

D. The movies that were most successful at the box office tended to have high average star ratings.

ID: a2b0fc3b



The share of the world's population living in cities has increased dramatically since 1970, but this change has not been uniform. France and Japan, for example, were already heavily urbanized in 1970, with 70% or more of the population living in cities. The main contributors to the world's urbanization since 1970 have been countries like Algeria, whose population went from _____

Which choice most effectively uses data from the graph to complete the assertion?

A. around 50% urban in 1970 to around 90% urban in 2020.

B. less than 40% urban in 1970 to around 90% urban in 2020.

C. less than 20% urban in 1970 to more than 50% urban in 2020.

D. around 40% urban in 1970 to more than 70% urban in 2020.

ID: e1546fd6

Average Nitrate and Phosphate Concentrations in Seawater after Volcanic Eruption

Nutrient	Seawater in lava-affected area, 5–45 meters below surface	Seawater in lava-affected area, 75–125 meters below surface	Seawater outside of lava-affected area, 5–45 meters below surface	Seawater outside of lava-affected area, 75–125 meters below surface
Nitrate (micromoles per liter)	3.1	0.4	≤0.03	≤0.01
Phosphate (micromoles per liter)	0.17	0.09	0.14	0.06

After a volcanic eruption spilled lava into North Pacific Ocean waters, a dramatic increase of diatoms (a kind of phytoplankton) near the surface occurred. Scientists assumed the diatoms were thriving on nutrients such as phosphate from the lava, but analysis showed these nutrients weren't present near the surface in forms diatoms can consume. However, there was an abundance of usable nitrate, a nutrient usually found in much deeper water and almost never found in lava. Microbial oceanographer Sonya Dyhrman and colleagues believe that as the lava plunged nearly 300 meters below the surface it dislodged pockets of this nutrient, releasing it to float upward, given that _____

Which choice most effectively uses data from the table to complete the statement?

A. at 5–45 meters below the surface, the average concentration of phosphate was about the same in the seawater in the lava-affected area as in the seawater outside of the lava-affected area.

B. for both depth ranges measured, the average concentrations of nitrate were substantially higher in the seawater in the lava-affected area than in the seawater outside of the lava-affected area.

C. for both depth ranges measured in the seawater in the lava-affected area, the average concentrations of nitrate were substantially higher than the average concentrations of phosphate.

D. in the seawater outside of the lava-affected area, there was little change in the average concentration of nitrate from 75–125 meters below the surface to 5–45 meters below the surface.

ID: 29cde5fa

"Mr. Cornelius Johnson, Office-Seeker" is a 1900 short story by Paul Laurence Dunbar. In the story, the narrator describes Mr. Cornelius Johnson's appearance as conveying his exaggerated sense of his importance: _____

Which quotation from "Mr. Cornelius Johnson, Office-Seeker" most effectively illustrates the claim?

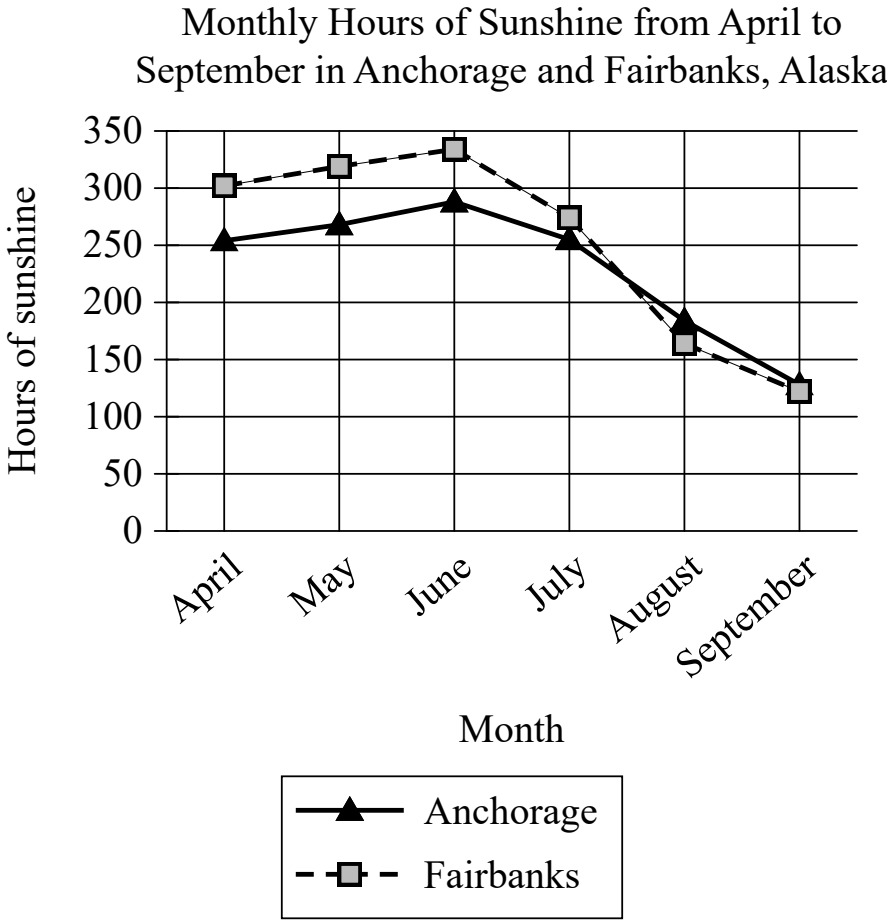
A. "He carried himself always as if he were passing under his own triumphal arch."

B. "The grey Prince Albert was scrupulously buttoned about his form, and a shiny top hat replaced the felt of the afternoon."

C. "It was a beautiful day in balmy May and the sun shone pleasantly on Mr. Cornelius Johnson's very spruce Prince Albert suit of grey as he alighted from the train in Washington."

D. "Mr. Cornelius Johnson always spoke in a large and important tone."

ID: 8a668840



A student is researching monthly hours of sunshine in different cities in Alaska. When comparing trends in Anchorage and Fairbanks, the student concludes that the two cities show a similar pattern in the monthly hours of sunshine from April to September.

Which choice best describes data from the graph that support the student's conclusion?

A. The monthly hours of sunshine in both Anchorage and Fairbanks hold steady in June and July before beginning to decline in August.

B. The monthly hours of sunshine in both Anchorage and Fairbanks increase from April to June and then decrease from June to September.

C. Anchorage and Fairbanks both have less than 200 monthly hours of sunshine from April to September.

D. Anchorage and Fairbanks both have more than 300 monthly hours of sunshine from April to June and less than 200 hours from July to September.

ID: c538954d

Sense and Sensibility is an 1811 novel by Jane Austen. In the novel, Austen describes Marianne Dashwood's ability to persuade others of the rightness of her artistic judgments, as is evident when Marianne visits with John Willoughby, a potential suitor: _____

Which quotation from *Sense and Sensibility* most effectively illustrates the claim?

A. "Above all, when she heard him declare, that of music and dancing he was passionately fond, she gave him such a look of approbation as secured the largest share of his discourse to herself for the rest of his stay."

B. "Their taste was strikingly alike. The same books, the same passages were idolized by each—or if any difference appeared, any objection arose, it lasted no longer than till the force of her arguments and the brightness of her eyes could be displayed."

C. "It was only necessary to mention any favourite amusement to engage her to talk. She could not be silent when such points were introduced, and she had neither shyness nor reserve in their discussion."

D. "They speedily discovered that their enjoyment of dancing and music was mutual, and that it arose from a general conformity of judgment in all that related to either. Encouraged by this to a further examination of his opinions, she proceeded to question him on the subject of books."

ID: 5fb6ed10

The Land of Enchantment is a 1906 travel book by Lilian Whiting. In the book, which describes the experience of traveling through the southwestern United States by train, Whiting reflects on the escape from everyday life that such a journey provides: _____

Which quotation from *The Land of Enchantment* most effectively illustrates the claim?

A. “The opportunities and advantages already offered and constantly increasing are greater than would at first be considered possible.”

B. “The social and the picturesque charm of the long journey is singularly enhanced by the leisurely stops made for refreshment.”

C. “The real journey begins, of course, at Chicago, and as these trains leave in the evening the traveller fares forth in the seclusion of his berth.”

D. “One experiences a certain sense of detachment from ordinary day and daylight duties that is exhilarating.”

ID: 0c622cfb

Although it's clear that Mars once had liquid water on its surface, astronomers have debated whether the evidence of ancient water reflects a prolonged phase of warm, wet conditions—the so-called wet and warm scenario—or a brief period of melting in an otherwise consistently frozen environment. Researchers Benjamin T. Cardenas and Michael P. Lamb recently added to this debate by using data from NASA and the Mars Orbiter Laser Altimeter to map the topography of what is now a large basin in Mars's northern hemisphere. Cardenas and Lamb concluded that the wet and warm scenario is likely correct.

Which finding about the basin, if true, would most directly support Cardenas and Lamb's conclusion?

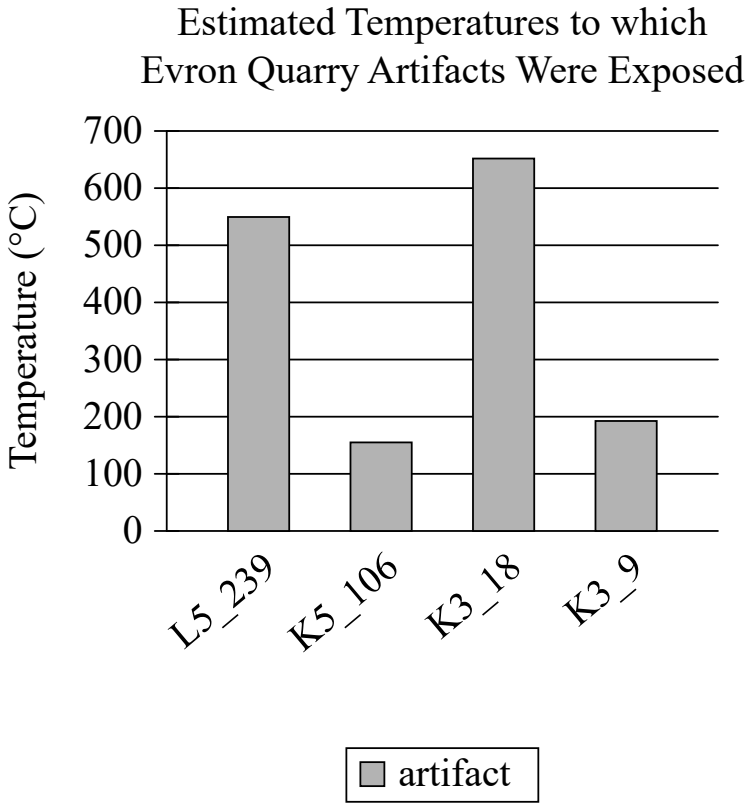
A. Its dimensions and shape indicate that it is unlikely to have formed as the result of an asteroid or comet impact.

B. It has features suggesting that it once held an ocean that underwent gradual sea-level changes over an extended time.

C. Its physical characteristics are most consistent with it having formed as a result of a massive but short-lived influx of liquid water.

D. It is surrounded by channels that could have been formed either by running water or by flowing lava.

ID: 7edfb2c5



Flint artifacts dating to 800,000 to 1,000,000 years ago have been recovered from the Evron Quarry in Israel. Likely created by the hominin *Homo erectus*, the artifacts have no visual features suggesting that they were exposed to fire, leading some scholars to conclude that these hominins had not acquired control of fire. But Zane Stepka and colleagues recently used a new method to determine whether these artifacts had been exposed to temperatures above 400°C (the typical temperature campfires reach) and concluded that the hominins who inhabited the site may have had control of fire.

Which choice best describes data in the graph that support the team's conclusion?

A. Artifacts K5_106 and K3_9 were exposed to temperatures above 400°C.

B. Artifacts L5_239 and K3_18 were exposed to temperatures of approximately 550°C and 650°C, respectively.

C. All of the artifacts were exposed to temperatures above 100°C.

D. Artifact K3_9 was exposed to a higher temperature than was artifact K5_106.

ID: 67b59a67

Plants like potatoes, tomatoes, and soybeans are susceptible to bacterial wilt disease caused by the bacteria *Ralstonia solanacearum*. A multinational team of scientists led by Zhong Wei studied whether other microbes in the soil might influence the degree to which plants are affected by the disease. The team sampled soil surrounding individual tomato plants over time and compared the results of plants that became diseased with those that remained healthy. They concluded that the presence of certain microbes in the soil might explain the difference between healthy and diseased plants.

Which finding, if true, would most directly support the team's conclusion?

A. The soil surrounding healthy plants contained significantly higher concentrations of microbes known to inhibit *Ralstonia solanacearum* than the soil surrounding diseased plants did.

B. The soil surrounding the plants contained high concentrations of *Ralstonia solanacearum* regardless of whether the plants were affected by wilt disease.

C. The soil surrounding healthy plants tended to have significantly higher moisture levels than the soil surrounding diseased plants did.

D. By the end of the experiment, over half the plants had been affected by wilt disease regardless of differences in the types and concentrations of microbes in the surrounding soil.

ID: 25176ff8

“Mrs. Spring Fragrance” is a 1912 short story by Sui Sin Far. In the story, Mrs. Spring Fragrance, a Chinese immigrant living in Seattle, is traveling in California. In letters to her husband and friend, she demonstrates her concern for what’s happening at her home in Seattle while she is away: _____

Which quotation from Mrs. Spring Fragrance’s letters most effectively illustrates the claim?

A. “My honorable cousin is preparing for the Fifth Moon Festival, and wishes me to compound for the occasion some American ‘fudge,’ for which delectable sweet, made by my clumsy hands, you have sometimes shown a slight prejudice.”

B. “Next week I accompany Ah Oi to the beauteous town of San José. There will we be met by the son of the Illustrious Teacher.”

C. “Forget not to care for the cat, the birds, and the flowers. Do not eat too quickly nor fan too vigorously now that the weather is warming.”

D. “I am enjoying a most agreeable visit, and American friends, as also our own, strive benevolently for the accomplishment of my pleasure.”

ID: dd72993d

Rivers rich in sediment appear yellow, while increases in red algae make rivers appear red. To track things like the sediment or algae content of large US rivers, John R. Gardner and colleagues used satellite data to determine the dominant visible wavelengths of light measured for various segments of these rivers. The researchers classified wavelengths of 495 nanometers (nm) and below as red, wavelengths between 495 and 560 nm as blue, and wavelengths of 560 nm and above as yellow. The researchers concluded that for the Missouri River, segments flowing into lakes tend to carry more sediment than those flowing out of lakes.

Which finding, if true, would most directly support the researchers' conclusion?

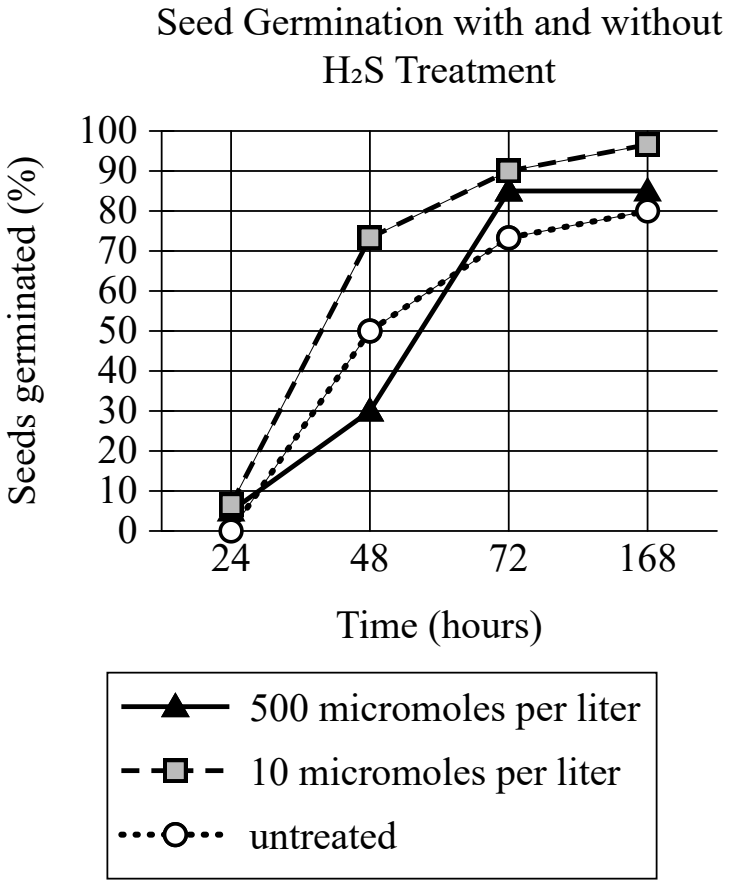
A. The segments of the Missouri River that had higher levels of chlorophyll-a, which contributes to the green color of photosynthetic organisms, have dominant wavelengths of light between 490 and 560 nm.

B. In lakes through which segments of the Missouri River pass, the dominant wavelength of light tended to be above 560 nm near the lakes' shores and below 560 nm in the lakes' centers.

C. The majority of the segments of the Missouri River were found to have dominant wavelengths of light significantly higher than 560 nm.

D. Segments of the Missouri River flowing into lakes typically had dominant wavelengths of light above 560 nm, while segments flowing out of lakes typically had dominant wavelengths below 560 nm.

ID: 8a584241



In high concentrations, hydrogen sulfide (H_2S) is typically toxic to many plants. Frederick D. Dooley and colleagues wanted to understand what effects low doses of H_2S might have on plant growth. They treated bean, corn, wheat, and pea seeds with various concentrations (measured in micromoles per liter) of H_2S and tracked the germination of those seeds along with the germination of untreated seeds. Treatment with particular concentrations of H_2S was associated with accelerated germination: for example, _____

Which choice most effectively uses data from the graph to complete the statement?

A. at 24 hours, less than 10% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas more than 90% of those seeds had germinated at 168 hours.

B. at 48 hours, more than 70% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 50% of untreated seeds had germinated.

C. at 168 hours, more than 90% of seeds treated with H_2S at concentrations of 10 or 500 micromoles per liter had germinated, whereas less than 70% of untreated seeds had germinated.

D. at 48 hours, approximately 50% of seeds treated with H_2S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 30% of untreated seeds had germinated.

ID: a13541c0

Sandra Cisneros's 1984 novella *The House on Mango Street* made a lasting impact on US literature. Its depiction of Mexican American culture inspired later authors to examine their own heritage within their fictional works. Also influential was the book's portrayal of the main character, Esperanza, during a pivotal year of her youth. This insightful depiction of a preteen girl encouraged authors who, like Cisneros herself, are Latina to use fictional works to examine experiences from their own youth.

Which statement, if true, would most strongly support the claim in the underlined sentence?

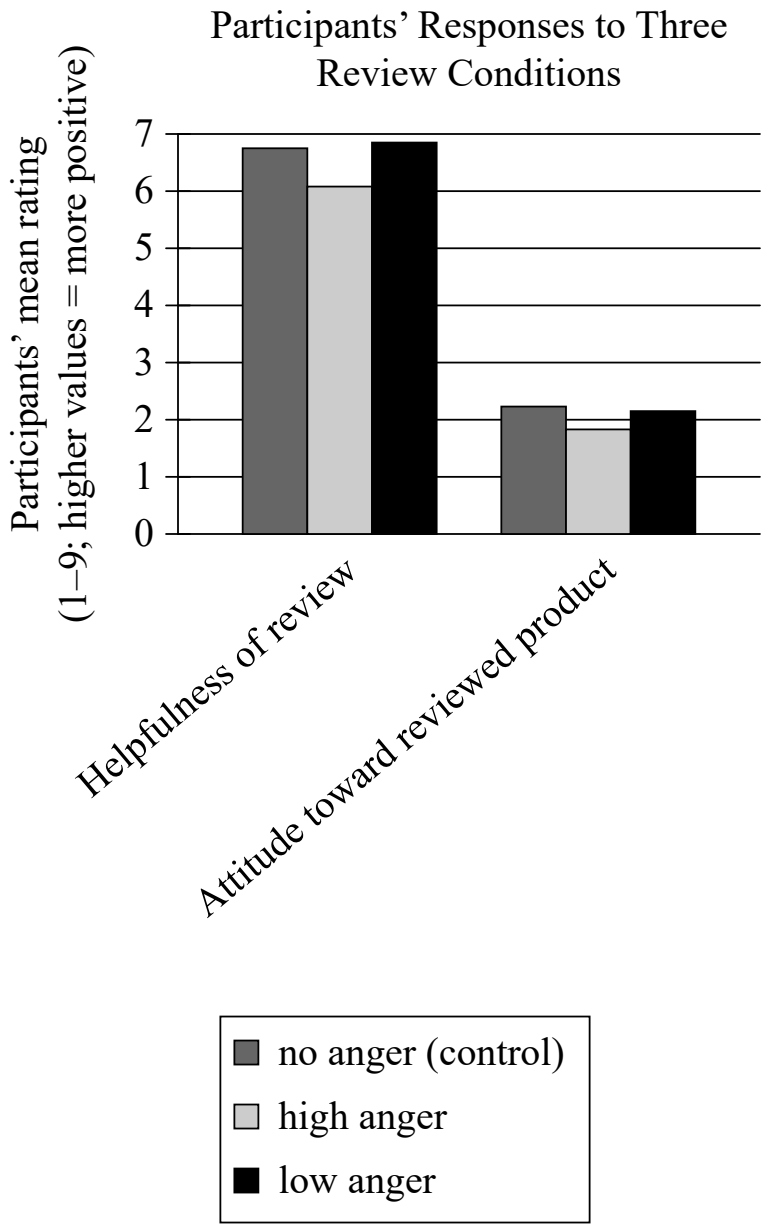
A. In interviews, a number of Latina authors say that *The House on Mango Street* inspired them to write about their own adolescence in their novels.

B. In published writings, several prominent authors who are not Latina say that reading *The House on Mango Street* influenced their approach to writing fiction.

C. The *House on Mango Street* has sold over six million copies and is one of the most commonly read books among high school and university students in the US.

D. Since 1984, new novels about young Latina characters by Latina authors have often been compared to *The House on Mango Street*.

ID: d5b9ed0d



To understand how expressions of anger in reviews of products affect readers of those reviews, business scholar Dezhi Yin and colleagues measured study participants' responses to three versions of the same negative review—a control review expressing no anger, a review expressing a high degree of anger, and a review expressing a low degree of anger. Reviewing the data, a student concludes that the mere presence of anger in a review may not negatively affect readers' perceptions of the review, but a high degree of anger in a review does worsen readers' perceptions of the review.

Which choice best describes data from the graph that support the students' conclusion?

A. On average, participants' ratings of the helpfulness of the review were substantially higher than were participants' ratings of the reviewed product regardless of which type of review participants had seen.

B. Compared with participants who saw the control review, participants who saw the low-anger review rated the review as slightly more helpful, whereas participants who saw the high-anger review rated the review as less helpful.

C. Participants who saw the low-anger review rated the review as slightly more helpful than participants who saw the control review did, but participants' attitude toward the reviewed product was slightly worse when participants saw the low-anger review than when they saw the no-anger review.

D. Compared with participants who saw the low-anger review, participants who saw the high-anger review rated the review as less helpful and had a less positive attitude toward the reviewed product.

ID: 7cbb9764

Accomplished printmaker and sculptor Elizabeth Catlett (1915–2012) used her art to explore the Black experience in the United States. In a paper for an art history class, a student claims that Catlett had a particular talent for unifying various artistic traditions and styles in her work.

Which quotation from a scholar describing Catlett's work would best support the student's claim?

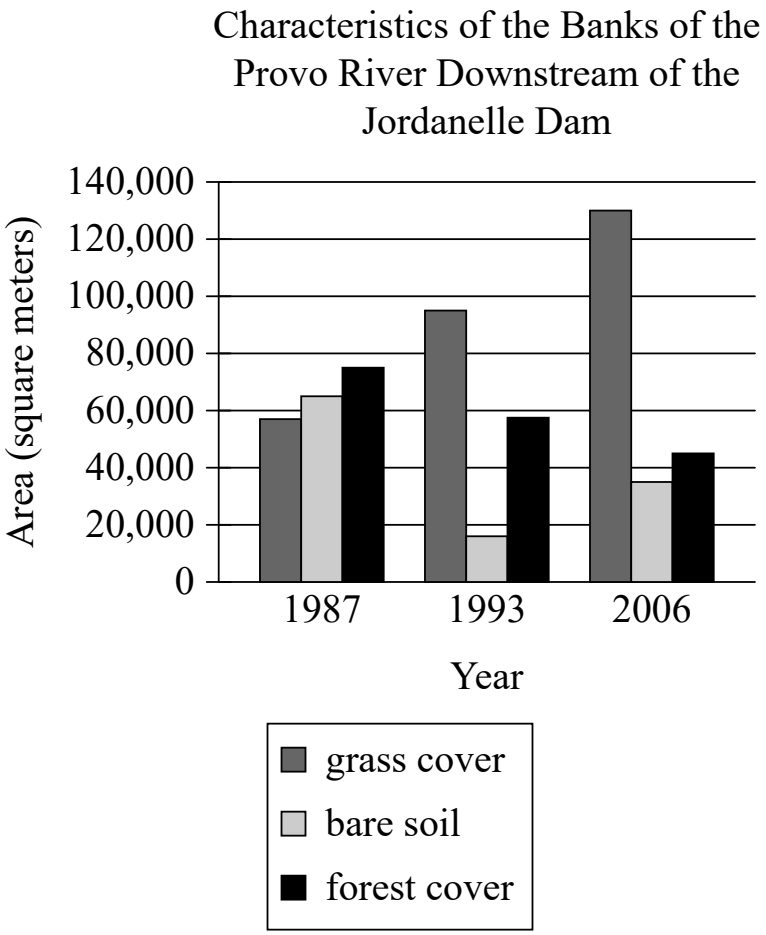
A. "In *Mother and Child*, a sculpture of two Black figures, Catlett used an ancient Indigenous sculpting technique and combined the visual aesthetic of modern Mexican muralists with that of German artist Kathe Kollwitz."

B. "In her collage *New Generation*, Catlett overlaid fabric onto the canvas to represent the clothing of a father and his toddler, positioned to evoke classic images of a mother and child."

C. "Created in 1968, Catlett's sculpture *Black Unity*, a stylized fist sculpted from mahogany and measuring two feet across, remains an important piece and has received renewed and well-deserved attention in recent years."

D. "One series of Catlett's prints, made by the artist using the linoleum cut method, depicts several notable African American women, including Harriet Tubman and Sojourner Truth."

ID: d83c3d54



The Jordanelle Dam was built on the Provo River in Utah in 1992. Earth scientist Adriana E. Martinez and colleagues tracked changes to the environment on the banks of the river downstream of the dam, including how much grass and forest cover were present. They concluded that the dam changed the flow of the river in ways that benefited grass plants but didn't benefit trees.

Which choice best describes data from the graph that support Martinez and colleagues' conclusion?

A. The lowest amount of grass cover was approximately 58,000 square meters, and the highest amount of forest cover was approximately 75,000 square meters.

B. There was more grass cover than forest cover in 1987, and this difference increased dramatically in 1993 and again in 2006.

C. There was less grass cover than bare soil in 1987 but more grass cover than bare soil in 1993 and 2006, whereas there was more forest cover than bare soil in all three years.

D. Grass cover increased from 1987 to 1993 and from 1993 to 2006, whereas forest cover decreased in those periods.

ID: 22e4d633

Although many transposons, DNA sequences that move within an organism's genome through shuffling or duplication, have become corrupted and inactive over time, those from the long interspersed nuclear elements (LINE) family appear to remain active in the genomes of some species. In humans, they are functionally important within the hippocampus, a brain structure that supports complex cognitive processes. When the results of molecular analysis of two species of octopus—an animal known for its intelligence—were announced in 2022, the confirmation of a LINE transposon in *Octopus vulgaris* and *Octopus bimaculoides* genomes prompted researchers to hypothesize that that transposon family is tied to a species' capacity for advanced cognition.

Which finding, if true, would most directly support the researchers' hypothesis?

A. The LINE transposon in *O. vulgaris* and *O. bimaculoides* genomes is active in an octopus brain structure that functions similarly to the human hippocampus.

B. The human genome contains multiple transposons from the LINE family that are all primarily active in the hippocampus.

C. A consistent number of copies of LINE transposons is present across the genomes of most octopus species, with few known corruptions.

D. *O. vulgaris* and *O. bimaculoides* have smaller brains than humans do relative to body size, but their genomes contain sequences from a wider variety of transposon families.

ID: 409058ee

To protect themselves when being attacked, hagfish—jawless marine animals that resemble eels—will release large quantities of slimy, mucus-like threads. Because these threads are unusually strong and elastic, scientist Atsuko Negishi and her colleagues have been trying to recreate them in a lab as an eco-friendly alternative to petroleum-based fibers that are often used in fabrics. The researchers want to reproduce the threads in the lab because farming hagfish for their slime would be expensive and potentially harmful to the hagfish.

Which choice best states the text's main idea?

A. The slimy threads that hagfish release might help researchers create a new kind of fabric.

B. Hagfish have inspired researchers to develop a new petroleum-based fabric.

C. Hagfish are not well suited to being raised in captivity.

D. The ability of hagfish to slime their attackers compensates for their being jawless.

ID: 95146ebb

The ice melted on a Norwegian mountain during a particularly warm summer in 2019, revealing a 1,700-year-old sandal to a mountaineer looking for artifacts. The sandal would normally have degraded quickly, but it was instead well preserved for centuries by the surrounding ice. According to archaeologist Espen Finstad and his team, the sandal, like those worn by imperial Romans, wouldn't have offered any protection from the cold in the mountains, so some kind of insulation, like fabric or animal skin, would have needed to be worn on the feet with the sandal.

What does the text indicate about the discovery of the sandal?

A. Temperatures contributed to both protecting and revealing the sandal.

B. The discovery revealed that the Roman Empire had more influence on Norway than archaeologists previously assumed.

C. Archaeologists would have found the sandal eventually without help from the general public.

D. The sandal would have degraded if it hadn't been removed from the ice.

ID: adbcbce0

The following text is adapted from Christina Rossetti's 1881 poem "Monna Innominata 2."

I wish I could remember that first day,
  First hour, first moment of your meeting me,
  If bright or dim the season, it might be
Summer or Winter for [all] I can say;
So unrecorded did it slip away,
  So blind was I to see and to foresee,
  So dull to mark the budding of my tree
That would not blossom yet for many a May.

Which choice best states the main idea of the text?

A. The speaker celebrates how the passage of time has strengthened a relationship that once seemed unimportant.

B. Because the speaker did not anticipate how important a relationship would become, she cannot recall how the relationship began, which she regrets.

C. As the anniversary of the beginning of an important relationship approaches, the speaker feels conflicted about how best to commemorate it.

D. After years of neglecting a once valuable relationship, the speaker worries it may be too late for her to salvage the relationship.

ID: ad680167

The recovery of a 1,000-year-old Chinese shipwreck in the Java Sea near present-day Indonesia has yielded a treasure trove of artifacts, including thousands of small ceramic bowls. Using a portable X-ray fluorescence analyzer tool, Lisa Niziolek and her team were able to detect the chemical composition of these bowls without damaging them. By comparing the chemical signatures of the bowls with those of the materials still at old Chinese kiln sites, Niziolek and her team can pinpoint which Chinese kilns likely produced the ceramic bowls.

Which choice best states the main idea of the text?

A. Because of a new technology, researchers can locate and recover more shipwrecks than they could in the past.

B. Researchers have been able to identify the location of a number of Chinese kilns in operation 1,000 years ago.

C. With the help of a special tool, researchers have determined the likely origin of bowls recovered from a shipwreck.

D. Before the invention of portable X-ray fluorescence, researchers needed to take a small piece out of an artifact to analyze its components.

ID: d5c2a4d4

The following text is adapted from Guy de Maupassant's nineteenth-century short story "The Trip of Le Horla" (translated by Albert M. C. McMaster, A. E. Henderson, Mme. Quesada, et al.). The narrator is part of a group traveling in a hot-air balloon at night.

> The earth no longer seems to exist, it is buried in milky vapors that resemble a sea. We are now alone in space with the moon, which looks like another balloon travelling opposite us; and our balloon, which shines in the air, appears like another, larger moon, a world wandering in the sky amid the stars, through infinity. We no longer speak, think nor live; we float along through space in delicious inertia. The air which is bearing us up has made of us all beings which resemble itself, silent, joyous, irresponsible beings, peculiarly alert, although motionless.

Which choice best states the main idea of the text?

A. The narrator feels a growing sense of isolation even though his companions are nearby during the balloon ride.

B. The narrator and his companions are completely absorbed in the change in perspective they gain while riding in the balloon.

C. The narrator and his companions are troubled by the disorienting effects of the altitude while riding in the balloon.

D. The narrator is pleasantly surprised by his companions' unrestrained enthusiasm about the sensation of riding in the balloon.

ID: 659c6c1d

The following text is adapted from Robert Louis Stevenson's 1883 novel *Treasure Island*. Bill is a sailor staying at the Admiral Benbow, an inn run by the narrator's parents.

Every day when [Bill] came back from his stroll he would ask if any seafaring men had gone by along the road. At first we thought it was the want of company of his own kind that made him ask this question, but at last we began to see he was desirous to avoid them. When a seaman did [stay] at the Admiral Benbow (as now and then some did) he would look in at him through the curtained door before he entered the parlour; and he was always sure to be as silent as a mouse when any such was present.

According to the text, why does Bill regularly ask about "seafaring men"?

A. He's hoping to find an old friend and fellow sailor.

B. He's trying to secure a job as part of the crew on a new ship.

C. He isn't sure that other guests at the inn will be welcoming of sailors.

D. He doesn't want to encounter any other sailor unexpectedly.

ID: 3f05e40f

In many of his sculptures, artist Richard Hunt uses broad forms rather than extreme accuracy to hint at specific people or ideas. In his first major work, *Arachne* (1956), Hunt constructed the mythical character Arachne, a weaver who was changed into a spider, by welding bits of steel together into something that, although vaguely human, is strange and machine-like. And his large bronze sculpture *The Light of Truth* (2021) commemorates activist and journalist Ida B. Wells using mainly flowing, curved pieces of metal that create stylized flame.

Which choice best states the text's main idea about Hunt?

A. He often depicts the subjects of his sculptures using an unrealistic style.

B. He uses different kinds of materials depending on what kind of sculpture he plans to create.

C. He tends to base his art on important historical figures rather than on fictional characters.

D. He has altered his approach to sculpture over time, and his works have become increasingly abstract.

ID: f8befe75

Many intellectual histories of the Black Power movement of the 1960s and 1970s rely heavily on essays and other explicitly ideological works as primary sources, a tendency that can overrepresent the perspectives of a small number of thinkers, most of whom were male. Historian Ashley D. Farmer has shown that expanding the array of primary sources to encompass more types of print material—including political cartoons, advertisements, and artwork—leads to a much better understanding of the movement and the crucial and diverse roles that Black women played in shaping it.

Which choice best describes the main idea of the text?

A. Farmer's methods and research have enriched the historical understanding of the Black Power movement and Black women's contributions to it.

B. Before Farmer's research, historians had largely ignored the intellectual dimensions of the Black Power movement.

C. Other historians of the Black Power movement have criticized Farmer's use of unconventional primary sources.

D. The figures in the Black Power movement whom historians tend to cite would have agreed with Farmer's conclusions about women's roles in the movement.

ID: 12030076

NASA's Aspera mission, led by Carlos Vargas, will investigate the circumgalactic medium (CGM), the huge swaths of low-density gas that fill and surround galaxies. Specifically, the team will focus on portions of the gas that exist in a "warm-hot" phase: these portions haven't previously been observable but are thought to fuel new star formation and hold most of the mass that makes up a galaxy. Using a telescope capable of revealing these parts of the CGM, the Aspera mission should help answer long-standing questions about how galaxies emerge, change, and even interact.

Which choice best states the main idea of the text?

A. As the leader of NASA's Aspera mission, Vargas will be the first person to investigate the makeup of the CGM.

B. Although galaxies that are surrounded by the CGM have been studied, researchers have been unable to directly observe low-density gas in the CGM in the "warm-hot" phase.

C. Researchers don't yet have a complete understanding of the process of galaxy evolution but have raised the possibility that galaxies interact with each other at times.

D. The Aspera mission is expected to produce the first direct observations of CGM gas in the "warm-hot" phase, which likely has an important role in the evolution of galaxies.

ID: a59245a1

The painter María Izquierdo played an important role in the development of twentieth-century Mexican art, but her work has never been well-known in the United States except among art historians. One reason for Izquierdo's relative obscurity is the enormous popularity of some of her peers. In particular, the painters Frida Kahlo and Diego Rivera have so captivated the interest of US audiences that Izquierdo and other Mexican artists from the period often get overlooked, despite the high quality of their work.

Which choice best states the main idea of the text?

A. Izquierdo's work is not as well-known in the United States as it should be because Kahlo and Rivera draw so much of the public's attention.

B. During Izquierdo's lifetime, her paintings were displayed in galleries in the United States much more frequently than paintings by Kahlo and Rivera were.

C. Izquierdo painted some of the same subjects that Kahlo and Rivera painted but used different techniques than they used.

D. Few of Izquierdo's works are in galleries today because she produced only a small number of paintings.

ID: dc3ea63e

To dye wool, Navajo (Diné) weaver Lillie Taylor uses plants and vegetables from Arizona, where she lives. For example, she achieved the deep reds and browns featured in her 2003 rug *In the Path of the Four Seasons* by using Arizona dock roots, drying and grinding them before mixing the powder with water to create a dye bath. To intensify the appearance of certain colors, Taylor also sometimes mixes in clay obtained from nearby soil.

Which choice best states the main idea of the text?

A. Reds and browns are not commonly featured in most of Taylor's rugs.

B. Taylor draws on local resources in the approach she uses to dye wool.

C. Taylor finds it difficult to locate Arizona dock root in the desert.

D. *In the Path of the Four Seasons* is widely acclaimed for its many colors and innovative weaving techniques.

ID: 5325b3cc

Philadelphia's Black Pearl Chamber Orchestra, founded by Jeri Lynne Johnson, performs classical music, from well-known compositions by Beethoven to contemporary works by Jessie Montgomery. For the orchestra's iConduct! program, Johnson invites community members to learn some basic elements of conducting and then experience conducting the Black Pearl orchestra themselves.

Which choice best states the main idea of the text?

A. The Black Pearl orchestra performs music from all over the world but mostly performs music composed by Philadelphians.

B. Johnson founded the Black Pearl orchestra to perform classical music by contemporary artist Jessie Montgomery.

C. The Black Pearl orchestra gives community members the chance to both listen to and participate in classical music performance.

D. Johnson has community members conduct an orchestra to demonstrate how difficult the task is.

ID: e677fa6c

The following text is adapted from Edgar Allan Poe's 1849 story "Landor's Cottage."

> During a pedestrian trip last summer, through one or two of the river counties of New York, I found myself, as the day declined, somewhat embarrassed about the road I was pursuing. The land undulated very remarkably; and my path, for the last hour, had wound about and about so confusedly, in its effort to keep in the valleys, that I no longer knew in what direction lay the sweet village of B——, where I had determined to stop for the night.

Which choice best states the main idea of the text?

A. The narrator remembers a trip he took and admits to getting lost.

B. The narrator recalls fond memories of a journey that he took through some beautiful river counties.

C. The narrator describes what he saw during a long trip through a frequently visited location.

D. The narrator explains the difficulties he encountered on a trip and how he overcame them.

ID: 11a9f635

Paleontologists searching for signs of ancient life have found many fossilized specimens of prehistoric human ancestors, including several from the Pleistocene era discovered in a geological formation in the Minatogawa quarry in Japan. However, to study the emergence of the earliest multicellular organisms to appear on Earth, researchers must turn elsewhere, such as to the Ediacaran geological formation at Mistaken Point in Canada. A UNESCO World Heritage Site, the 146-hectare reserve contains more than 10,000 fossils that together document a critical moment in evolutionary history.

What does the text indicate about the geological formation at Mistaken Point?

A. It holds a greater number of fossils but from a smaller variety of species than the formation in the Minatogawa quarry does.

B. It has provided evidence that the earliest human species may have emerged before the Pleistocene era.

C. It is widely considered by paleontologists to be the most valuable source of information about prehistoric life forms.

D. It contains specimens from an older time period than those found in the formation in the Minatogawa quarry.

ID: 602b47c7

Biologists have predicted that birds' feather structures vary with habitat temperature, but this hadn't been tested in mountain environments. Ornithologist Sahas Barve studied feathers from 249 songbird species inhabiting different elevations—and thus experiencing different temperatures—in the Himalaya Mountains. He found that feathers of high-elevation species not only have a greater proportion of warming downy sections to flat and smooth sections than do feathers of low-elevation species, but high-elevation species' feathers also tend to be longer, providing a thicker layer of insulation.

Which choice best states the main idea of the text?

A. Barve's investigation shows that some species of Himalayan songbirds have evolved feathers that better regulate body temperature than do the feathers of other species, contradicting previous predictions.

B. Barve found an association between habitat temperature and feather structure among Himalayan songbirds, lending new support to a general prediction.

C. Barve discovered that songbirds have adapted to their environment by growing feathers without flat and smooth sections, complicating an earlier hypothesis.

D. The results of Barve's study suggest that the ability of birds to withstand cold temperatures is determined more strongly by feather length than feather structure, challenging an established belief.

ID: 87aa7bab

A common assumption among art historians is that the invention of photography in the mid-nineteenth century displaced the painted portrait in the public consciousness. The diminishing popularity of the portrait miniature, which coincided with the rise of photography, seems to support this claim. However, photography's impact on the portrait miniature may be overstated. Although records from art exhibitions in the Netherlands from 1820 to 1892 show a decrease in the number of both full-sized and miniature portraits submitted, this trend was established before the invention of photography.

Based on the text, what can be concluded about the diminishing popularity of the portrait miniature in the nineteenth century?

A. Factors other than the rise of photography may be more directly responsible for the portrait miniature's decline.

B. Although portrait miniatures became less common than photographs, they were widely regarded as having more artistic merit.

C. The popularity of the portrait miniature likely persisted for longer than art historians have assumed.

D. As demand for portrait miniatures decreased, portrait artists likely shifted their creative focus to photography.

ID: d748c3fd

In her 2021 article "Throwaway History: Towards a Historiography of Ephemera," scholar Anne Garner discusses John Johnson (1882–1956), a devoted collector of items intended to be discarded, including bus tickets and campaign pamphlets. Johnson recognized that scholarly institutions considered his expansive collection of ephemera to be worthless—indeed, it wasn't until 1968, after Johnson's death, that Oxford University's Bodleian Library acquired the collection, having grasped the items' potential value to historians and other researchers. Hence, the example of Johnson serves to _____

Which choice most logically completes the text?

A. demonstrate the difficulties faced by contemporary historians in conducting research at the Bodleian Library without access to ephemera.

B. represent the challenge of incorporating examples of ephemera into the collections of libraries and other scholarly institutions.

C. lend support to arguments by historians and other researchers who continue to assert that ephemera holds no value for scholars.

D. illustrate both the relatively low scholarly regard in which ephemera was once held and the later recognition of ephemera's possible utility.

ID: 3190835d

Some businesses believe that when employees are interrupted while doing their work, they experience a decrease in energy and productivity. However, a team led by Harshad Puranik, who studies management, has found that interruptions by colleagues can have a social component that increases employees' sense of belonging, resulting in greater job satisfaction that benefits employees and employers. Therefore, businesses should recognize that _____

Which choice most logically completes the text?

A. the interpersonal benefits of some interruptions in the workplace may offset the perceived negative effects.

B. in order to maximize productivity, employers should be willing to interrupt employees frequently throughout the day.

C. most employees avoid interrupting colleagues because they don't appreciate being interrupted themselves.

D. in order to cultivate an ideal workplace environment, interruptions of work should be discouraged.

ID: b1fab3e1

Violins made by Antonio Stradivari and other craftspeople in the sixteenth to eighteenth centuries in Cremona, Italy, produce a sound that is considered superior to that of modern stringed instruments. Some experts have claimed that the type of wood used to create Cremonese violins is responsible for their prized sound, but modern and Cremonese violins are made of the same kinds of wood: maple and spruce. New analysis, however, has revealed unique indications that the wood in the older violins was chemically treated by the makers, leading researchers to suggest that _____

Which choice most logically completes the text?

A. Cremonese violins probably were not considered superior to other instruments at the time they were made.

B. the sound quality of Cremonese violins results in part from a method the craftspeople used to alter the wood.

C. if modern violins were made of a wood other than maple or spruce, they likely would sound as good as Cremonese violins.

D. the current process of making violins is the same process that was used centuries ago by Cremonese craftspeople.

ID: cac82f9b

Biologist Natacha Bodenhausen and colleagues analyzed the naturally occurring bacterial communities associated with leaves and roots of wild *Arabidopsis thaliana*, a small flowering plant. The researchers found many of the same bacterial genera in both the plants' leaves and roots. To explain this, the researchers pointed to the general proximity of *A. thaliana* leaves to the ground and noted that rain splashing off soil could bring soil-based bacteria into contact with the leaves. Alternatively, the researchers noted that wind, which may be a source of bacteria in the aboveground portion of plants, could also bring bacteria to the soil and roots. Either explanation suggests that _____

Which choice most logically completes the text?

A. bacteria carried by wind are typically less beneficial to *A. thaliana* than soil-based bacteria are.

B. some bacteria in *A. thaliana* leaves and roots may share a common source.

C. many bacteria in *A. thaliana* leaves may have been deposited by means other than rain.

D. *A. thaliana* leaves and roots are especially vulnerable to harmful bacteria.

ID: bcf2f169

Ana Castillo's 1986 novel *The Mixquiahuala Letters* is a story told entirely through expressive letters from the narrator to her friend—letters that Castillo suggests could be read in several different orders. As they began reading it in class, some students remarked that they found the novel's letter format daunting and its treatment of gender relations old-fashioned. The professor, however, pointed out that the novel is written in modern-sounding language and addresses issues that still matter today, suggesting that *The Mixquiahuala Letters* _____

Which choice most logically completes the text?

A. has more to say about gender relations than other novels from the same period.

B. is more relevant to contemporary audiences than it may seem at first.

C. is easier to read than many contemporary novels that focus on friendship.

D. is best understood after multiple readings in different orders.

ID: 299c5303

As the name suggests, dramaturges originated in theater, where they continue to serve a variety of functions: conducting historical research for directors, compiling character biographies for actors, and perhaps most importantly, helping writers of plays and musicals to hone the works' stories and characters. Performance scholar Susan Manning observes that many choreographers, like playwrights and musical theater writers, are concerned with storytelling and characterization. In fact, some choreographers describe the dances they create as expressions of narrative through movement; it is therefore unsurprising that _____

Which choice most logically completes the text?

A. dramaturges can have a profound impact on the artistic direction of plays and musicals.

B. choreographers developing dances with narrative elements frequently engage dramaturges to assist in refining those elements.

C. dances by choreographers who incorporate narrative elements are more accessible to audiences than dances by choreographers who do not.

D. some directors and actors rely too heavily on dramaturges to complete certain research tasks.

ID: 485962a6

Astronomers investigated the Arabia Terra region of Mars because it appears to contain irregularly shaped craters that may have been caused by massive volcanic explosions. In their investigations of Arabia Terra, the researchers found remnants of ash deposits in an amount and thickness that would result from a massive volcanic eruption. However, erosion and past resurfacing events could have modified the surface of the planet. Therefore, _____

Which choice most logically completes the text?

A. the current makeup of the Arabia Terra region might not accurately reflect the volcanic activity of Mars's past.

B. eruptions from Mars's volcanoes were likely not as massive as astronomers previously believed.

C. ash was most likely expelled from multiple different volcanoes on Mars's surface.

D. the craters found in the Arabia Terra region were necessarily created by events other than volcanic eruptions.

ID: 575e67df

By running computer simulations of the development of our solar system, André Izidoro, Rajdeep Dasgupta, and colleagues concluded that the Sun may have been surrounded by three giant dust rings before the planets started to form. The researchers suggest that the materials in the innermost ring became the four planets closest to the Sun, the materials in the middle ring produced the rest of the planets, and the materials in the outermost ring created the asteroids and other small bodies in the region beyond Neptune. In one simulation, the researchers delayed the initial formation of the middle ring, causing oversized super-Earths to begin developing from the innermost ring. The researchers therefore hypothesize that _____

Which choice most logically completes the text?

A. the middle ring formed earlier in the solar system's development than the initial simulations suggested.

B. the timing of the initial formation of the middle ring played an important role in determining the eventual size of Earth.

C. if the formation of the outermost ring had occurred earlier in a simulation, all the planets would have become super-Earths.

D. the innermost ring actually formed into all the planets in our solar system, not just the four closest to the Sun.

ID: 350e2336

The Haitian Declaration of Independence was issued in 1804, bringing to an end the revolution against colonial France that began in 1791. Written in French, which was not the first language of most Haitians but which was used throughout Europe as the language of international diplomacy, the declaration notes that Haiti will not bring rebellion to other Caribbean nations, promises to respect the sovereignty of its neighbors—widely understood as a reassurance to the United States—and sets up Haiti as an example for future struggles against colonizers (an implicit reference to the many colonies then found in the Americas). So even though the declaration is explicitly addressed to the Haitian people, it's reasonable to conclude that _____

Which choice most logically completes the text?

A. aspects of the declaration were modeled on similar documents from other countries.

B. the French government may have been surprised by the declaration.

C. many Haitian people opposed the revolution and the declaration.

D. the declaration actually had several intended audiences.

ID: 3882ddf6

To investigate the history of plate subduction—when one of Earth's tectonic plates slides beneath another—Sarah M. Aarons and colleagues compared ancient rocks from the Acasta Gneiss Complex in Canada to modern rocks. Using isotope analysis, the researchers found that Acasta rocks dating to about 4.02 billion years ago (bya) most strongly resemble modern rocks formed in a plume setting (an area in which hot rocks from Earth's mantle flow upward into the crust). By contrast, they found that Acasta rocks dating to about 3.75 bya and 3.6 bya have an isotope composition that is similar to that of modern rocks formed in a subduction setting. Aarons's team therefore concluded that _____

Which choice most logically completes the text?

A. subduction-like processes began occurring in some locations no later than 3.75 bya.

B. subduction replaced mantle plume formation as the most common geological process by about 4.02 bya.

C. the majority of the rocks in the Acasta Gneiss Complex formed through subduction.

D. the rocks in the Acasta Gneiss Complex are of a more recent origin than scientists previously thought.

ID: 4e9afd7a

The Indus River valley civilization flourished in South Asia from 3300 BCE to 1300 BCE. Many examples of the civilization's writing system exist, but researchers haven't yet deciphered it or identified which ancient language it represents. Nevertheless, archaeologists have found historical artifacts, such as clay figures and jewelry, that provide information about the civilization's customs and how its communities were organized. The archaeologists' findings therefore suggest that _____

Which choice most logically completes the text?

A. investigating an ancient civilization is easier without knowledge of the civilization's language.

B. knowing an ancient civilization's language isn't necessary in order to learn details about the civilization.

C. archaeological research should focus on finding additional artifacts rather than deciphering ancient languages.

D. examining the civilization's historical artifacts has resolved the debate about this civilization's language.

ID: 5432d1de

It's common for jazz musicians and fans to refer to certain songs as having "swing," indicating that the songs provoke a strong feeling, like the impulse to tap one's foot or dance. The exact acoustic properties that give a song swing, however, have long been thought to be undefinable. To investigate swing, a team led by physicist Corentin Nelias delayed the downbeats and synchronized the offbeats in jazz piano solos and asked jazz musicians to compare the intensity of swing in each modified piece with the intensity of swing in the original piece. They found that participants were more than seven times likelier to characterize the modified songs as having swing than to characterize the original versions as having swing, suggesting that _____

Which choice most logically completes the text?

A. synchronized offbeats tend to give a song swing regardless of whether downbeats are delayed.

B. the acoustic properties that give a song swing are not easy for jazz musicians to manipulate.

C. jazz songs that feature the piano are more likely to have swing than are jazz songs that do not feature the piano.

D. the timing of downbeats and offbeats may play a crucial role in giving a song swing.

ID: 8fbed1cb

When the Vinland Map, a map of the world purported to date to the mid-1400s, surfaced in 1957, some scholars believed it demonstrated that European knowledge of the eastern coast of present-day North America predated Christopher Columbus's 1492 arrival. In 2021, a team including conservators Marie-France Lemay and Paula Zyats and materials scientist Anikó Bezur performed an extensive analysis of the map and the ink used. They found that the ink contains titanium dioxide, a compound that was first introduced in ink manufacturing in the early 1900s. Therefore, the team concluded that _____

Which choice most logically completes the text?

A. mid-1400s Europeans could not have known about the eastern coast of present-day North America.

B. the Vinland Map could not have been drawn by mid-1400s mapmakers.

C. mapmakers must have used titanium compounds in their ink in the 1400s.

D. there isn't enough information to determine when the ink was created.

ID: c95995bc

Colonized by Spain in the 1600s, New Mexico is home to a dialect of Spanish that differs significantly from dialects spoken in Spain's other former colonies in the Americas. Most notably, the New Mexican dialect retains older features of the language that other dialects lost in later centuries. But why would it have done so? New Mexico was so distant from population centers in Spain's other colonies that it attracted few colonists after its initial colonization. Geographical isolation in turn would have limited the exposure of New Mexican colonists to changes occurring to Spanish grammar and vocabulary elsewhere in the empire. Thus, the present-day uniqueness of the New Mexican dialect suggests the extent to which _____

Which choice most logically completes the text?

A. a language can protect itself from being influenced by other languages.

B. the grammar and vocabulary of any given language change from one generation to the next.

C. geographical isolation can influence how a language develops.

D. speakers of one dialect of a language can understand speakers of another dialect of that language.

ID: 55688b3c

Dutch painters in the sixteenth and seventeenth centuries often showed tables filled with large wheels of cheese or carved shards of butter. Some art historians, noting that dairy products were a major component of the Dutch diet, interpret these depictions as reflections of everyday Dutch eating habits. However, a group of researchers recently reviewed hundreds of food-related paintings and found that lemons—which could only be acquired in the Netherlands at great cost, since they had to be imported from warmer climates—feature in Dutch paintings of the period more than three times as frequently as dairy products do, thereby casting doubt on the idea that _____

Which choice most logically completes the text?

A. dairy products were a more significant component of the Dutch diet of the period than lemons were.

B. food was a more popular subject among Dutch painters than it was among painters from other countries at the time.

C. depictions of food in Dutch paintings of the period should be taken as realistic representations of Dutch eating habits.

D. Dutch painters of the period may have depicted foods for symbolic reasons rather than to show what Dutch people typically ate.

ID: db876fd5

Songbirds learn to respond to and imitate their species' songs from an early age. With each generation, small differences are introduced that result in distinct variations—called dialects—among geographically isolated populations of the same species. A research study examined whether twelve-day-old *Ficedula hypoleuca* (pied flycatcher) nestlings prefer local dialects over the unfamiliar dialects of nonlocal *F. hypoleuca* populations: the more begging calls the nestlings made in response to a song, the stronger their preference. The researchers found that nestlings produced more begging calls in response to their own dialect than to nonlocal dialects. Since song preference plays a role in songbird mate selection, the finding suggests that _____

Which choice most logically completes the text?

A. *F. hypoleuca* nestlings' preference for their own dialect likely disappears as they mature to promote socialization between different *F. hypoleuca* populations.

B. *F. hypoleuca* nestlings who show an early preference for their own dialect are likely to receive more food from their caretakers than nestlings who show no preferences among any *F. hypoleuca* dialects.

C. *F. hypoleuca* nestlings' preference for their own dialect likely drives them when they mature to reproduce with other *F. hypoleuca* from local rather than nonlocal populations.

D. *F. hypoleuca* nestlings show a preference for both local *F. hypoleuca* dialects and the songs of other local songbirds over the songs of nonlocal birds of any species.